1 — Introduction to WAN Technologies

Theory: WAN vs. LAN — key differences and use cases. MPLS basics. Practical: Observe MPLS routing in a pre-configured network.

1. Comparison table: WAN vs LAN, with real-world examples

Aspect	LAN (Local Area Network)	WAN (Wide Area Network)
Geographic coverage	Covers a small area — one building or campus (e.g. IRCTC's internal office network in one location).	Covers large distances, connecting sites across cities/countries (e.g. Zomato's app talking to servers across different data centers).
Ownership / links used	Usually owned and managed entirely by one organisation using its own cabling.	Usually leases links from ISPs/telecom providers (e.g. Instagram traffic travelling over ISP backbone links between your phone and Meta's data centers).
Speed & cost	Very high speed, low cost per connection (Gigabit Ethernet is common).	Comparatively slower and more expensive per Mbps, since it depends on carrier infrastructure over long distances.

2. Diagram: Ahmedabad user connecting to a Flipkart data center in Bangalore over a WAN

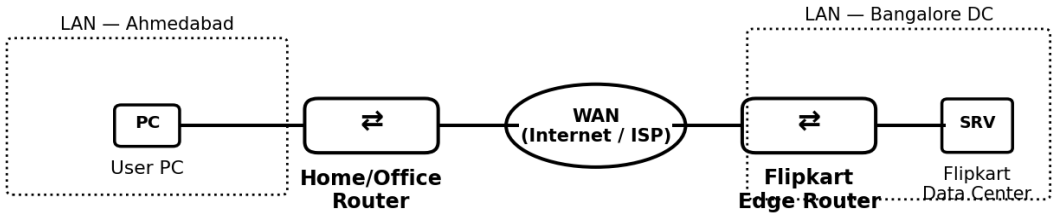


Fig: LAN segments connected across a WAN (illustrative)

The two dotted boxes are the LAN segments (the user's home/office network in Ahmedabad, and Flipkart's internal data-center network in Bangalore); the cloud in the middle is the WAN — the ISP/internet path connecting the two LANs together.

3. Summary: what MPLS is and how it improves routing compared to traditional IP routing

MPLS (Multiprotocol Label Switching) is a WAN forwarding technique that sits between Layer 2 and Layer 3. Instead of every router along the path doing a full IP routing-table lookup for each packet, the ingress router (LER) attaches a short, fixed-length label to the packet. Core routers (LSRs) then forward the packet purely by looking up that label — a much faster operation than a full IP lookup — and swap it for a new label at each hop. This is called a Label-Switched Path (LSP). Because the path is pre-determined by labels rather than recalculated hop-by-hop, MPLS also makes it easy to guarantee quality of service (QoS) and traffic-engineer specific paths for specific traffic types, something traditional hop-by-hop IP routing cannot do as efficiently.

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4. Observing the pre-configured MPLS network — two routing behaviors different from standard IP routing

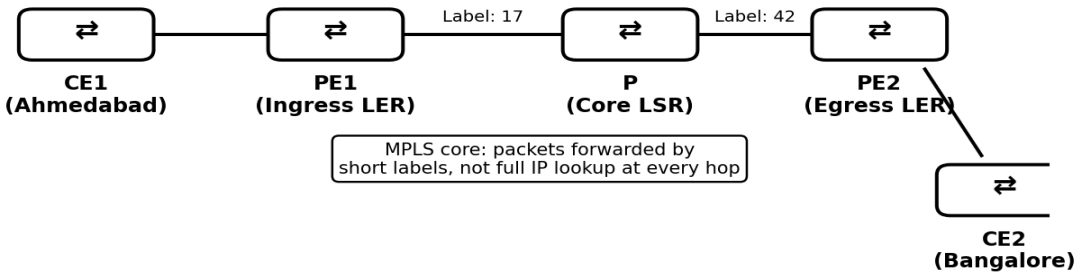


Fig: MPLS label-switched path across a WAN core (illustrative)

1. Forwarding by label, not IP address: on the core (P) router, show mpls forwarding-table shows packets being switched based on an incoming/outgoing label pair rather than a destination IP lookup — the IP header isn't inspected hop-by-hop inside the MPLS core.
2. Labels change at every hop: the same packet carries label 17 between PE1 and P, but that label is swapped to label 42 between P and PE2 (label swapping) — very different from IP routing, where the destination address in the header never changes end-to-end.

5. Three benefits of MPLS over traditional routing for high-traffic apps (YouTube, BookMyShow)

1. Faster forwarding at scale: label lookups are simpler than full IP routing-table lookups, so core routers carrying huge volumes of YouTube video traffic can forward packets with less per-hop delay.
2. Traffic engineering / guaranteed QoS: MPLS can steer latency-sensitive traffic (e.g. BookMyShow's live seat-booking requests during a big release) over a specific low-latency path, rather than leaving it to standard best-effort IP routing.
3. Reduced link congestion through explicit paths: MPLS can spread traffic across multiple paths deliberately (instead of all traffic converging on the one 'shortest' IP path), which helps avoid a single link becoming a bottleneck during traffic spikes like a popular video going viral.

2 — Introduction to VPNs

Theory: Types of VPNs — site-to-site, remote access. Practical: Configure a basic site-to-site VPN in Packet Tracer.

1. Two branch networks, each with its own router and basic IP addressing

Branch	Router LAN interface	PC IP	Public-facing interface
Branch-A (Ahmedabad)	Gig0/0: 10.1.1.1/24	PC-A: 10.1.1.10/24	Gig0/1: 203.0.113.1/30
Branch-B (Surat)	Gig0/0: 10.2.2.1/24	PC-B: 10.2.2.10/24	Gig0/1: 198.51.100.1/30

2. Configure a site-to-site VPN using GRE tunnel interfaces

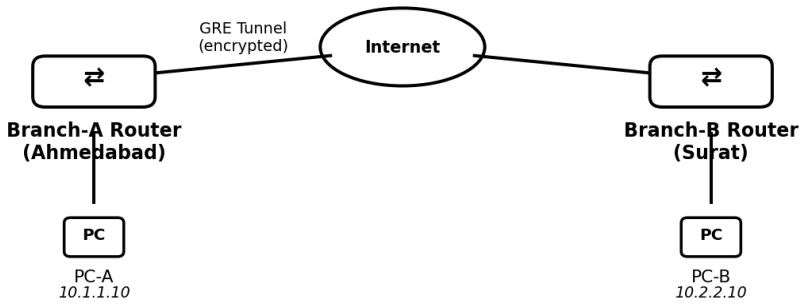


Fig: Site-to-site VPN between two branch routers (illustrative)

```
Branch-A(config)# interface tunnel 0
Branch-A(config-if)# ip address 172.16.0.1 255.255.255.252
Branch-A(config-if)# tunnel source 203.0.113.1
Branch-A(config-if)# tunnel destination 198.51.100.1
Branch-A(config)# ip route 10.2.2.0 255.255.255.0 tunnel 0

Branch-B(config)# interface tunnel 0
Branch-B(config-if)# ip address 172.16.0.2 255.255.255.252
Branch-B(config-if)# tunnel source 198.51.100.1
Branch-B(config-if)# tunnel destination 203.0.113.1
Branch-B(config)# ip route 10.1.1.0 255.255.255.0 tunnel 0
```

After adding these routes pointing at the tunnel interface, PC-A (10.1.1.10) can ping PC-B (10.2.2.10) across the GRE tunnel.

3. Simulate a remote-access VPN using Packet Tracer's built-in VPN client

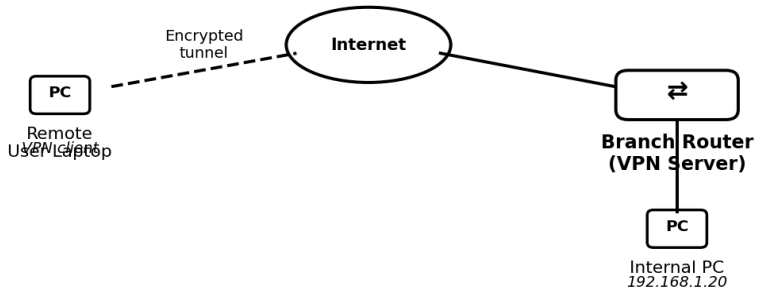


Fig: Remote-access VPN — client connects directly to branch router (illustrative)

On the remote PC's Desktop > VPN tab, set: VPN Type = PPTP (or supported client type), Server IP = the branch router's public interface, plus a Username/Password matching a local user configured on the router. Connecting brings up a tunnel icon on the PC and lets it reach internal addresses like 192.168.1.20 behind the branch router.

4. Site-to-site VPN vs remote-access VPN — differences, advantages, disadvantages

	Site-to-site VPN	**Remote-access VPN**
What it connects	Two fixed networks/branch offices to each other, always-on.	A single remote user's device directly into a network, on demand.
Advantages	• Transparent to end users — no client software needed on every PC • Good for constant branch-to-branch traffic	• Very flexible — any single traveling/remote employee can connect from anywhere • No need to pre-configure a tunnel for every possible remote user
Disadvantages	• Not practical for many individual roaming users • Requires a static/known public IP at both ends	• Requires VPN client software/config on every user device • Typically slower / more overhead per-user than a dedicated site link

5. Step-by-step site-to-site VPN configuration commands used

```
! Branch-A
interface tunnel 0
 ip address 172.16.0.1 255.255.255.252
 tunnel source 203.0.113.1
 tunnel destination 198.51.100.1
ip route 10.2.2.0 255.255.255.0 tunnel 0

! Branch-B
interface tunnel 0
 ip address 172.16.0.2 255.255.255.252
 tunnel source 198.51.100.1
 tunnel destination 203.0.113.1
ip route 10.1.1.0 255.255.255.0 tunnel 0
```

3 — NAT and PAT Concepts

Theory: What is NAT/PAT? Importance in IP address conservation. Practical: Configure NAT and PAT on a router.

1. Static NAT between an internal PC (private IP) and an external PC (public IP)

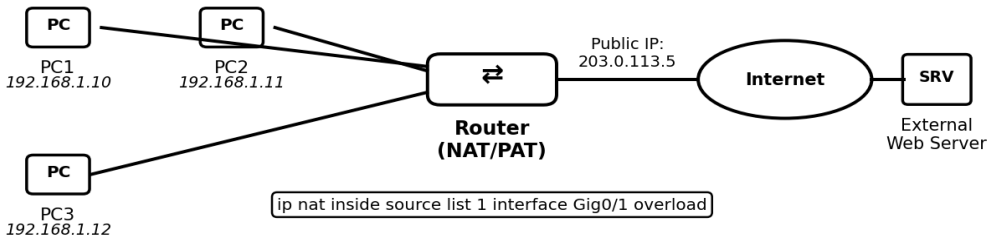


Fig: PAT — multiple private IPs sharing one public IP (illustrative)

```
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config)# interface gig0/1
Router(config-if)# ip address 203.0.113.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config)# ip nat inside source static 192.168.1.10 203.0.113.10
```

2. Configure PAT so two internal PCs share a single public IP

```
Router(config)# access-list 1 permit 192.168.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface gig0/1 overload
```

3. Test with a ping and capture the NAT translation table

```
Router# show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
icmp 203.0.113.1:1024   192.168.1.10:1024  203.0.113.10:1024  203.0.113.10:1024
```

4. Add a third internal PC, update NAT/PAT, verify all three appear in the table

```
Router# show ip nat translations
Pro  Inside global        Inside local       Outside local     Outside global
icmp 203.0.113.1:1024     192.168.1.10:1024  ...               ...
icmp 203.0.113.1:2048     192.168.1.11:2048  ...               ...
icmp 203.0.113.1:3072     192.168.1.12:3072  ...               ...
```

Because PAT (with overload) was already used rather than plain static NAT, no configuration change was actually required to support the third PC — PAT tracks each internal host by a unique port number behind the same single public IP, so it scales to additional internal hosts automatically.

5. How NAT/PAT help apps like Zomato or Flipkart serve millions of users with limited public IPs

A company like Flipkart doesn't need a separate public IP address for every internal server or employee device — PAT lets an entire office or data-center segment share a small number of public IPs, since each internal host is distinguished by port number rather than by IP address. This massively conserves the limited pool of public IPv4 addresses while still letting every internal device reach the internet.

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Two real-world scenarios where incorrect NAT/PAT setup could break connectivity:

1. If the NAT 'inside'/'outside' interface roles are configured backwards (or missing on one interface), the router will not translate addresses at all, and any external user trying to reach a Flipkart service will simply see the request time out.
2. If the access-list used by 'ip nat inside source list' is written too narrowly (e.g. missing a newer internal subnet added when the office expanded), devices on that subnet will not get translated and will be unable to reach the internet at all — a silent scaling gap in the NAT config.

4 — Weekly Practice and Troubleshooting

Practical: Simulate a WAN setup with VPN and NAT. Task: Troubleshoot connectivity issues in the setup. Checkpoint: Quiz on WAN technologies, VPNs, and NAT/PAT.

1. Simulate a WAN connecting two branches via VPN tunnel, with NAT on the edge routers

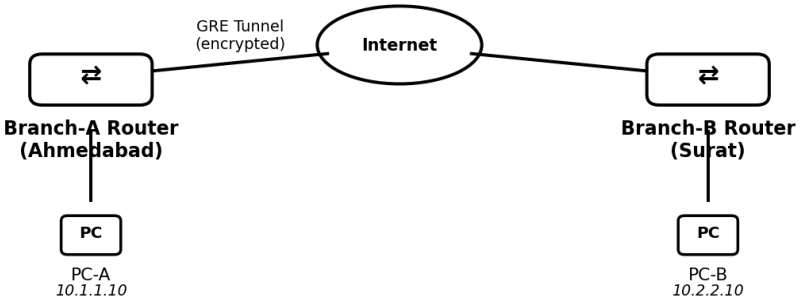


Fig: Site-to-site VPN between two branch routers (illustrative)

```
! Edge router NAT (example on Branch-A), alongside the GRE tunnel already configured:
Branch-A(config)# ip nat inside source list 1 interface gig0/1 overload
Branch-A(config)# access-list 1 permit 10.1.1.0 0.0.0.255
```

2. Configure PAT so three internal devices browse the internet via one public IP

```
Router(config)# access-list 1 permit 10.1.1.0 0.0.0.255
Router(config)# ip nat inside source list 1 interface gig0/1 overload
```

3. Intentionally misconfigure the VPN pre-shared key, then diagnose and fix it

```
Router# show crypto isakmp sa
dst             src             state          conn-id  status
198.51.100.1    203.0.113.1     MM_NO_STATE    1        ACTIVE   <- stuck, PSK mismatch

Router# show run | include key
crypto isakmp key WRONGKEY123 address 198.51.100.1
```

An MM_NO_STATE result stuck at the first Main Mode message (rather than reaching QM_IDLE) is the classic symptom of a pre-shared key mismatch. The fix is to re-enter matching keys on both routers:

```
Router(config)# no crypto isakmp key WRONGKEY123 address 198.51.100.1
Router(config)# crypto isakmp key CorrectSharedKey456 address 198.51.100.1
```

4. Troubleshoot: one branch can reach the internet but not the other branch via VPN

Diagnostic order used:

- show ip route — checked whether a route to the remote branch's subnet exists at all, pointing at the tunnel interface (found it missing on one router).
- show interfaces tunnel 0 — confirmed the tunnel interface itself was up/up (it was).
- show run | section nat — checked that the NAT access-list wasn't accidentally also matching (and translating) traffic destined for the other branch, which would break the VPN traffic by NAT'ing it before it reached the tunnel.

Root cause found: the static route for the remote branch's subnet via the tunnel interface was missing, so return traffic had no path back through the VPN and fell through to the default route (out to the internet) instead. Adding the missing 'ip route <remote-subnet> <mask> tunnel 0' resolved it.

5. Peer review: swap configuration with a classmate, find and fix one misconfiguration

Using show and debug commands only (not guessing): ran show ip interface brief to confirm all interfaces were up, show ip nat translations to confirm NAT was actively creating entries, and show crypto isakmp sa / show crypto ipsec sa to confirm the VPN phase 1 and phase 2 were both established. In the classmate's file, phase 1 was up but phase 2 (show crypto ipsec sa) showed zero packets encrypted — tracked down to a mismatched access-list defining 'interesting traffic' for the VPN, which was corrected so it exactly matched on both routers.

Inter-VLAN Routing Concepts — Router-on-a-Stick

Theory: Why is inter-VLAN routing needed? VLANs cannot communicate with each other by default; the solution is a router or Layer 3 switch. Methods covered: Router-on-a-Stick (single router interface, sub-interfaces per VLAN) and Layer 3 switch routing. Practical: Router-on-a-Stick configuration — enable router sub-interfaces for VLANs.

1. Create three VLANs (Insta, Zomato, Flipkart) on a single switch, one PC each

```
Switch(config)# vlan 10
Switch(config-vlan)# name Insta
Switch(config)# vlan 20
Switch(config-vlan)# name Zomato
Switch(config)# vlan 30
Switch(config-vlan)# name Flipkart
Switch(config)# interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config)# interface fa0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config)# interface fa0/3
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 30
```

2. Router-on-a-Stick: sub-interfaces for Insta, Zomato, Flipkart

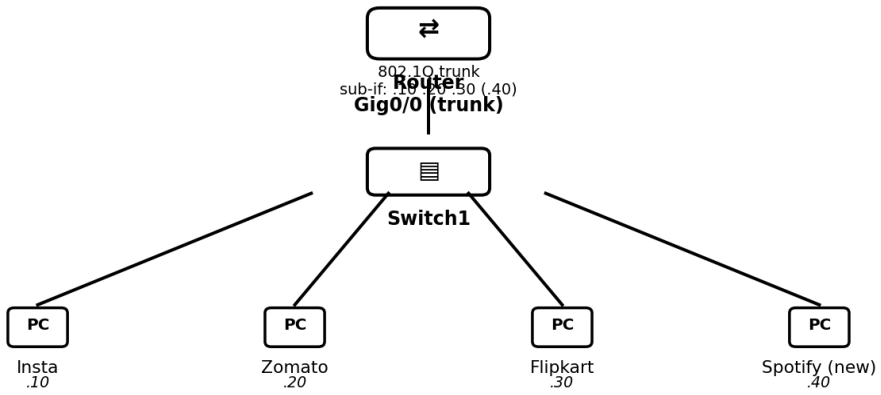


Fig: Router-on-a-Stick serving four VLANs (illustrative)

```
Router(config)# interface gig0/0
Router(config-if)# no shutdown
Router(config)# interface gig0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config)# interface gig0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config)# interface gig0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

3. Trunk port to the router, verify with show interfaces trunk

```
Switch(config)# interface fa0/4
Switch(config-if)# switchport trunk encapsulation dot1q
Switch(config-if)# switchport mode trunk
```

```
Switch# show interfaces trunk
Port      Mode    Encapsulation  Status       Native vlan
Fa0/4     on      802.1q         trunking     1
```

4. Test inter-VLAN connectivity: ping from Insta PC to Zomato PC

With the sub-interfaces and trunk correctly configured, and each PC's default gateway set to its own VLAN's sub-interface IP (e.g. the Insta PC uses 192.168.10.1), the ping from Insta to Zomato succeeds: the frame leaves the Insta PC tagged (by the switch) as VLAN 10, crosses the trunk to the router, is routed by the Gig0/0.10↔Gig0/0.20 sub-interfaces, and is re-tagged as VLAN 20 on the way back down to the Zomato PC. If the ping instead failed, the most likely causes would be a PC's gateway not matching its sub-interface IP, a missing 'no shutdown' on the physical interface, or the trunk not actually carrying both VLANs.

5. Add a new VLAN 'Spotify' without disrupting existing VLANs

```
! On the switch:
Switch(config)# vlan 40
Switch(config-vlan)# name Spotify
Switch(config)# interface fa0/5
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 40

! On the router — add a new sub-interface, existing ones are untouched:
Router(config)# interface gig0/0.40
Router(config-subif)# encapsulation dot1Q 40
Router(config-subif)# ip address 192.168.40.1 255.255.255.0
```

Because each VLAN has its own independent sub-interface, adding Gig0/0.40 for Spotify has no effect on the existing Gig0/0.10/.20/.30 sub-interfaces — Insta, Zomato and Flipkart continue to communicate exactly as before.

Layer 3 Switch Inter-VLAN Routing

Theory: Using a Layer 3 switch — no need for a router; uses SVIs (Switch Virtual Interfaces) for routing; faster than Router-on-a-Stick. Practical: Configuring inter-VLAN routing on a Layer 3 switch — enable IP routing.

1. Create three VLANs (Sales, Support, Marketing) on a Layer 3 switch

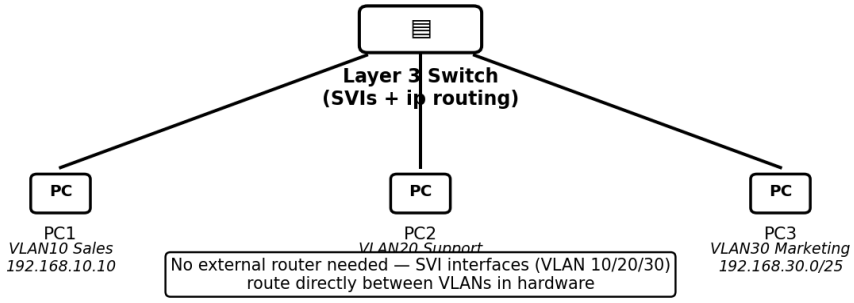


Fig: Inter-VLAN routing using SVIs on a Layer 3 switch (illustrative)

```
L3Switch(config)# vlan 10
L3Switch(config-vlan)# name Sales
L3Switch(config)# vlan 20
L3Switch(config-vlan)# name Support
L3Switch(config)# vlan 30
L3Switch(config-vlan)# name Marketing
L3Switch(config)# interface fa0/1
L3Switch(config-if)# switchport mode access
L3Switch(config-if)# switchport access vlan 10
```

2. Configure SVIs for each VLAN

```
L3Switch(config)# interface vlan 10
L3Switch(config-if)# ip address 192.168.10.1 255.255.255.0
L3Switch(config-if)# no shutdown
L3Switch(config)# interface vlan 20
L3Switch(config-if)# ip address 192.168.20.1 255.255.255.0
L3Switch(config-if)# no shutdown
L3Switch(config)# interface vlan 30
L3Switch(config-if)# ip address 192.168.30.1 255.255.255.0
L3Switch(config-if)# no shutdown
```

3. Enable IP routing on the Layer 3 switch

```
L3Switch(config)# ip routing
```

Without this single global command, a Layer 3 switch will still create the SVIs and let each VLAN's own devices talk to their SVI gateway, but it will not route traffic between different VLANs — 'ip routing' is what turns on the actual routing function between the SVIs.

4. Verify inter-VLAN routing: ping between a VLAN 10 PC and a VLAN 20 PC

```
PC1> ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:
Reply from 192.168.20.10: bytes=32 time=1ms TTL=254
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Reply from 192.168.20.10: bytes=32 time<1ms TTL=254
Ping statistics: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

5. Refactor VLAN 30 to use a /25 subnet mask instead of /24

```
L3Switch(config)# interface vlan 30
L3Switch(config-if)# ip address 192.168.30.1 255.255.255.128
```

Changing VLAN 30's SVI to a /25 mask (255.255.255.128) halves its address space to 126 usable hosts (192.168.30.1–192.168.30.126), and any device on that VLAN using an address in the upper half (192.168.30.128 onward) would now be on a different subnet and unable to reach the SVI gateway directly. This change is entirely local to VLAN 30 — it does not affect VLAN 10 or VLAN 20's own /24 addressing, and inter-VLAN routing between all three continues to work as long as every device within VLAN 30 is renumbered to fit inside the new, smaller /25 range.